

1 Time-Dependent Hamiltonians

To recap: the expectation value of an observable $\hat{A}$ at time t is given by

$$\langle \hat{A} \rangle_{\psi(t)} = \langle \psi(t) | \hat{A} | \psi(t) \rangle = \langle \psi(0) | \hat{U}^\dagger(t) \hat{A} \hat{U}(t) | \psi(0) \rangle = \langle \psi(0) | \hat{A}'(t) | \psi(0) \rangle = \langle \hat{A}'(t) \rangle_{\psi(0)}, \quad (1)$$

where $\hat{A}'(t) = \hat{U}^\dagger(t) \hat{A} \hat{U}(t)$ is the operator in the Heisenberg picture.

- **Schrödinger Picture:** States evolve in time,

$$|\psi(t)\rangle_S = \hat{U}(t) |\psi(0)\rangle, \quad \hat{A}_S. \quad (2)$$

- **Heisenberg Picture:** Operators evolve in time,

$$|\psi(t)\rangle_H = |\psi(0)\rangle = \hat{U}^\dagger(t) |\psi(t)\rangle_S, \quad \hat{A}_H(t) = \hat{U}^\dagger(t) \hat{A}_S \hat{U}(t). \quad (3)$$

Note that the operator in the Schrödinger picture does not evolve in time, but it can still be time-dependent. For simplicity, we omit the time dependence from our notation, but in principle one can write $\hat{A}_S(t)$.

When studying time-dependent perturbation theory, we will make use of a hybrid version of these: the interaction picture, where the operators evolve with one “free” part of the Hamiltonian, and the states evolve with the remaining interaction part (called the perturbation). What is important is that the connection to physical observables, such as expectation values, remains the same in all prescriptions.

We also saw the Schrödinger equation in the Schrödinger picture, governs the time evolution of the state

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle, \quad (4)$$

of the time evolution operator

$$i\hbar \frac{d}{dt} \hat{U}(t, t_0) = \hat{H}(t) \hat{U}(t, t_0), \quad \hat{U}(t_0, t_0) = \mathbb{1}. \quad (5)$$



Prove that

$$\frac{d}{dt} \langle \hat{A} \rangle = \frac{\langle [\hat{A}, \hat{H}] \rangle}{i\hbar} + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle, \quad (6)$$

where the last term is present if $\hat{A}$ has explicit time dependence. Show that this form holds in both the Schrödinger and Heisenberg pictures.

If the Hamiltonian is time-dependent, the time evolution operator is actually more complicated. We write it as the **time-ordered exponential**:

$$\hat{U}(t) = \mathcal{T} \exp \left(-\frac{i}{\hbar} \int_0^t \hat{H}(t') dt' \right), \quad (7)$$

where $\mathcal{T}$ denotes the time-ordering operator (defined later). Time ordering is necessary because, in general, the Hamiltonian at different times may not commute, $[\hat{H}(t_1), \hat{H}(t_2)] \neq 0$ for $t_1 \neq t_2$ (we interpret this case a little more in Section 1.1). So, the time evolution operator must account for the order in which the Hamiltonians are applied. If the Hamiltonian is time-dependent, but

its evolution does not care about the order of things (i.e., if $[\hat{H}(t_1), \hat{H}(t_2)] = 0$ for all t_1, t_2), then time ordering is irrelevant and we get

$$\hat{U}(t) = \exp \left(-\frac{i}{\hbar} \int_0^t \hat{H}(t') dt' \right). \quad (8)$$

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The result in Eq. (7) is given in S&N Chapter 2.1.2 and the derivation deferred to Chapter 5. You can find a sketch of the derivation in Section 1.2, where we introduce the *Dyson Series*. Solutions with time-ordered exponentials can be complicated and may require numerical methods. The Dyson series shows up in perturbation theory in quantum mechanics and in quantum field theory.

1.1 Non-Commuting Hamiltonians

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To understand the case where $[\hat{H}(t_1), \hat{H}(t_2)] \neq 0$, we will need two basic facts:

- 1) the expectation value of an operator that does not commute with the Hamiltonian is not conserved in time,
- 2) if two operators commute, they share a common set of eigenstates. In other words, they can be simultaneously diagonalized.

Therefore, if the Hamiltonians at different times do not commute, it means that energy is not conserved in our system (due to contact with an external bath or some external varying magnetic field, for example). In addition, the energy eigenstates of the system may be changing with time (in the Schrödinger picture). Note that the energy eigenstates are also changing when $[\hat{H}(t_1), \hat{H}(t_2)] = 0$ (including in the time-independent case), but only by a phase factor. The point is that in the most general case, the energy eigenstates at a future time may not just be a phase-evolved basis, but rather, “directions” may change too. This is equivalent to saying that a vector in the new eigenbasis evolves to be a mix of initial ones.

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Non-commutativity and the eigenstates: Using

$$\hat{H}(t) |n(t)\rangle = E_n(t) |n(t)\rangle, \quad (9)$$

show that the n -th energy eigenstate acquires a component along an orthogonal (assume also non-degenerate) energy eigenstate $|m(t)\rangle$ due to the time dependence of the Hamiltonian. In other words, prove that

$$\langle m(t) | \dot{n}(t) \rangle = -\frac{\langle m(t) | [\hat{H}(t), \partial_t \hat{H}(t)] | n(t) \rangle}{(E_n(t) - E_m(t))^2}, \quad m \neq n. \quad (10)$$

where $|m(t)\rangle$ and $|n(t)\rangle$ are instantaneous eigenstates of the Hamiltonian $\hat{H}(t)$ with eigenvalues $E_m(t) \neq E_n(t)$ and $m \neq n$. Note that for a infinitesimal step,

$$|n(t + dt)\rangle \simeq |n(t)\rangle + dt |\dot{n}(t)\rangle. \quad (11)$$

1.1.1 Time-Dependent driver in two-level system example

To see this in action, let us consider our two-level system with a time-dependent Hamiltonian given by

$$\hat{H}(t) = \frac{\hbar\omega_0}{2} \hat{\sigma}_z + V(t) \hat{\sigma}_x = \hat{H}_0 + \hat{H}_1(t), \quad (12)$$

where $\hat{\sigma}_x$ and $\hat{\sigma}_z$ are the Pauli matrices, ω_0 is a constant frequency, and $V(t)$ is a time-dependent potential. We separated the Hamiltonian into a time-independent and a time-dependent part explicitly. The Pauli matrices do not commute, i.e., $[\hat{\sigma}_z, \hat{\sigma}_x] = 2i\hat{\sigma}_y$ (cyclic permutation gives me a + sign), so the Hamiltonians at different times do not commute either unless $V(t_1) = V(t_2)$ (e.g. V is constant):

$$\begin{aligned} [\hat{H}(t_1), \hat{H}(t_2)] &= [\hat{H}_0 + \hat{H}_1(t_1), \hat{H}_0 + \hat{H}_1(t_2)] = [\hat{H}_0, \hat{H}_1(t_2)] + [\hat{H}_1(t_1), \hat{H}_0] \\ &= \frac{\hbar\omega_0}{2}(V(t_1) - V(t_2))[\hat{\sigma}_x, \hat{\sigma}_z] \\ &= i\hbar\omega_0(V(t_2) - V(t_1))\hat{\sigma}_y, \end{aligned} \quad (13)$$

Note how only the cross terms between the time-independent and time-dependent commutators survive. Taking $t_2 = t + dt$ and $t_1 = t$, we get the infinitesimal form

$$[\partial_t \hat{H}(t), \hat{H}(t)] = -i\hbar\omega_0 \dot{V}(t) \hat{\sigma}_y \quad (14)$$

We also have the time derivative of the Hamiltonian that can be explicitly calculated (check it indeed gives the same result above):

$$\partial_t \hat{H}(t) = \dot{V}(t) \hat{\sigma}_x. \quad (15)$$

With these two ingredients we can estimate the change in the energy of the system. It will turn out that we will need to work at order $(dt)^2$ to get the full picture.

Let's start with the rate of change of E :

$$\frac{dE}{dt} = \frac{d \langle \hat{H} \rangle_{\psi(t)}}{dt} = \langle \partial_t \hat{H}(t) \rangle_{\psi(t)} = \dot{V}(t) \langle \hat{\sigma}_x \rangle_{\psi(t)}, \quad (16)$$

noting that we used the Heisenberg equation of motion but the commutator term vanishes because it involves $[\hat{H}(t), \hat{H}(t)] = 0$. Anticipating its need, we also compute the second derivative. Now the commutator does not vanish because $[\hat{H}(t), \partial_t \hat{H}(t)] \neq 0$ in general:

$$\frac{d^2 E}{dt^2} = \frac{d}{dt} \langle \partial_t \hat{H}(t) \rangle_{\psi(t)} = \frac{1}{i\hbar} \langle [\partial_t \hat{H}(t), \hat{H}(t)] \rangle_{\psi(t)} + \langle \partial_t^2 \hat{H}(t) \rangle_{\psi(t)} \quad (17)$$

$$= -\omega_0 \dot{V}(t) \langle \hat{\sigma}_y \rangle_{\psi(t)} + \ddot{V}(t) \langle \hat{\sigma}_x \rangle_{\psi(t)}. \quad (18)$$

Taylor expanding the energy difference around a given time t , we get

$$dE = E(t + dt) - E(t) = \frac{dE}{dt} dt + \frac{1}{2} \frac{d^2 E}{dt^2} (dt)^2 + \mathcal{O}((dt)^3) \quad (19)$$

$$= \langle \partial_t \hat{H}(t) \rangle_{\psi(t)} dt + \frac{1}{2} \left(\langle \partial_t^2 \hat{H}(t) \rangle_{\psi(t)} + \frac{1}{i\hbar} \langle [\partial_t \hat{H}(t), \hat{H}(t)] \rangle_{\psi(t)} \right) (dt)^2 \quad (20)$$

Collecting all our results

$$dE \simeq \dot{V}(t) \langle \hat{\sigma}_x \rangle_{\psi(t)} dt + \frac{1}{2} \left(-\omega_0 \dot{V}(t) \langle \hat{\sigma}_y \rangle_{\psi(t)} + \ddot{V}(t) \langle \hat{\sigma}_x \rangle_{\psi(t)} \right) (dt)^2 \quad (21)$$

With some mathematical audacity, we can treat $dt \sim \Delta t$ as a small interval and treat $\dot{V} dt \simeq \Delta V$ as the change in the driving potential over that interval. This allows us to write a more intuitive form for the energy shift:

$$\Delta E \simeq \Delta V \left(\langle \hat{\sigma}_x \rangle_{\psi(t)} - \frac{\omega_0 \Delta t}{2} \langle \hat{\sigma}_y \rangle_{\psi(t)} \right) + \frac{\Delta V^2}{2} \langle \hat{\sigma}_x \rangle_{\psi(t)}. \quad (22)$$

The first bracket is really the most important contribution. The last term is there, but it is typically less consequential, it only corrects a contribution along the $\hat{\sigma}_x$ direction that already appears in the first term. The middle term, proportional to $\omega_0 \Delta t$, shifts the energy for a component in the orthogonal direction to the driving potential. Note that it comes from the non-commutativity of the Hamiltonian at different times.

Let us consider a specific state at the time t , $|\psi(t)\rangle = \frac{1}{\sqrt{2}}(|+\rangle + i|-\rangle)$, the eigenstate of $\hat{\sigma}_y$ with eigenvalue $+1$. In the spin- $\frac{1}{2}$ picture, the spin initially points exactly along the $\hat{y}$ direction. Then $\langle \hat{\sigma}_x \rangle_{\psi(t)} = \langle \hat{\sigma}_z \rangle_{\psi(t)} = 0$, and by Eq. (22)

$$\Delta E \simeq -\frac{1}{2}\omega_0 \Delta V(t) \Delta t \simeq -\frac{1}{2}\omega_0 \dot{V}(t) (\Delta t)^2. \quad (23)$$

Note that to see the energy shift in this specific state, we had to go to higher order in the small time interval Δt . Physically, the shift above is easy to understand. The drive only acts on the spin along the x direction, so initially it has no effect on a state pointing exactly along the $\hat{y}$ direction. However, the spin precesses around the $\hat{z}$ axis due to $\hat{H}_0$ (see S&N Chapter 2 or our two-level system notes). The interplay between the external drive and the intrinsic precession of the spin is manifest in the non-commutativity of the Hamiltonian.

1.2 Time-Ordered Evolution and the Dyson Series

In what follows, we give a brief derivation of Eq. (7) (well... as brief as I could make it). We go back to our differential form for the time evolution

$$\hat{U}(t + dt, t) = \hat{\mathbb{1}} - \frac{i}{\hbar} \hat{H}(t) dt + \mathcal{O}(dt^2).$$

with $H(t)$ evaluated at time t . From now on, we will not assume anything about $\hat{H}(t)$ other than that it is hermitian. As before, even without assuming commutativity, we can build up the full time evolution operator by composing many infinitesimal steps:

$$\hat{U}(t = t_N, t_0) = \hat{U}(t_N, t_{N-1}) \hat{U}(t_{N-1}, t_{N-2}) \cdots \hat{U}(t_1, t_0). \quad (24)$$

It is important that the earliest time evolutions appear on the rightmost side of this product. Using our infinitesimal form for $U(t + dt, t)$ for uniform steps $\Delta t = t_{k+1} - t_k$, we have

$$\hat{U}(t = t_N, t_0) = \left(\hat{\mathbb{1}} - \frac{i}{\hbar} \hat{H}(t_{N-1}) \Delta t + \dots \right) \cdots \left(\hat{\mathbb{1}} - \frac{i}{\hbar} \hat{H}(t_1) \Delta t + \dots \right) \left(\hat{\mathbb{1}} - \frac{i}{\hbar} \hat{H}(t_0) \Delta t + \dots \right), \quad (25)$$

Multiplying these out gives us terms of several orders in Δt (the lowest order being $(\Delta t)^0$ and highest order being $(\Delta t)^N$). Let us start collecting terms order by order and take the continuum limit $N \rightarrow \infty$ (with $N \times \Delta t$ fixed), also order by order. For convenience, let us define:

$$t_N \xrightarrow{N \rightarrow \infty} t_\infty = t. \quad (26)$$

Order $(\Delta t)^0$: There is only one such term, which is simply the identity operator:

$$\hat{\mathbb{1}} \xrightarrow{N \rightarrow \infty} \hat{\mathbb{1}}. \quad (27)$$

Order $(\Delta t)^1$: There are N such terms, each coming from picking one factor of $-\frac{i}{\hbar} \hat{H}(t_k) \Delta t$ from one of the N factors in Eq. (25), and $\hat{\mathbb{1}}$ from all the others. So, the sum of all order $(\Delta t)^1$ terms is

$$-\frac{i}{\hbar} \sum_{k=0}^{N-1} \hat{H}(t_k) \Delta t \xrightarrow{N \rightarrow \infty} -\frac{i}{\hbar} \int_{t_0}^t dt' \hat{H}(t'). \quad (28)$$

Note that t'_1 is a dummy integration variable now (not to be confused with the discrete time steps t_k). For a given t , the result above is a simple integral from t_0 to t . (Note that k is counting the number of intervals, so we count it from 0 to $N - 1$ here.)

Order $(\Delta t)^2$: There are two kinds of terms at this order. First and most important are those from multiplying $\mathcal{O}(\Delta t)$ terms from different $\hat{U}$ factors in Eq. (25). We find a total of

$$\binom{N}{2} = \frac{N(N-1)}{2}, \quad (29)$$

of such terms, which we can write as

$$\left(-\frac{i}{\hbar}\right)^2 \sum_{k_2 > k_1=0}^{N-1} \hat{H}(t_{k_2}) \hat{H}(t_{k_1}) (\Delta t)^2 \xrightarrow{N \rightarrow \infty} \left(-\frac{i}{\hbar}\right)^2 \int_{t_0}^t dt'_2 \int_{t_0}^{t'_2} dt'_1 \hat{H}(t'_2) \hat{H}(t'_1), \quad (30)$$

Note that each Hamiltonian factor must appear at the same time or at decreasing time orders since the earliest time evolutions appear on the rightmost side of the product in Eq. (25). For a fixed time t , we are integrating over all possible triangular regions in the t'_1 - t'_2 plane such that $t_0 < t'_1 < t'_2 < t$.

The second kind of contribution at $\mathcal{O}(\Delta t)^2$ comes from picking a term of $\mathcal{O}(\Delta t)^2$ from a single $\hat{U}$ factor. Clearly, there must be only a total of N of such terms, which we can write as

$$\frac{1}{2!} \left(-\frac{i}{\hbar}\right)^2 \sum_{k=0}^{N-1} \hat{H}(t_k) \hat{H}(t_k) (\Delta t)^2 = \frac{1}{2!} \left(-\frac{i}{\hbar}\right)^2 \sum_{k_1, k_2=0}^{N-1} (\Delta t \hat{H}(t_{k_1})) (\Delta t \hat{H}(t_{k_2})) \delta_{k_1 k_2} \xrightarrow{N \rightarrow \infty} 0. \quad (31)$$

One might be tempted to say that the continuum limit ought to recover something like a double integral over t_1 and t_2 with a delta function enforcing $t_1 = t_2$, something like

$$\int_{t_0}^t dt'_2 \int_{t_0}^{t'_2} dt'_1 \hat{H}(t'_2) \hat{H}(t'_1) \delta(t'_2 - t'_1).$$

However, this is not the case. You can convince yourself by just noting that the delta function has units of 1/time, while the Kronecker delta $\delta_{k_1 k_2}$ is dimensionless. The correct continuum limit that does recover the delta function would be

$$\frac{\delta_{k_1 k_2}}{\Delta t} \rightarrow \delta(t_{k_1} - t_{k_2}), \quad (32)$$

which is clearly not what we have in Eq. (31). Therefore, this second kind of contribution at order $(\Delta t)^2$ actually vanishes in the continuum limit. It corresponds to the very edge (the hypotenuse) of the triangular integration region, which has measure zero in the two-dimensional integral. Therefore, we can safely ignore it.

Order $(\Delta t)^n$: The pattern established above continues for higher orders in Δt . The leading number of contributions comes from picking n different Δt factors from n different $\hat{U}$ factors in Eq. (25), which gives us a total number of terms equal to

$$\binom{N}{n} = \frac{N!}{n!(N-n)!} \xrightarrow{N \rightarrow \infty} \frac{N^n}{n!}. \quad (33)$$

The sum of all such leading terms is

$$\begin{aligned} & \left(-\frac{i}{\hbar}\right)^n \sum_{k_n > \dots > k_1=0}^{N-1} \hat{H}(t_{k_n}) \hat{H}(t_{k_{n-1}}) \dots \hat{H}(t_{k_1}) (\Delta t)^n \\ & \xrightarrow{N \rightarrow \infty} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt'_n \int_{t_0}^{t'_n} dt'_{n-1} \dots \int_{t_0}^{t'_2} dt'_1 \hat{H}(t'_n) \hat{H}(t'_{n-1}) \dots \hat{H}(t'_1). \end{aligned} \quad (34)$$

We can say that for a fixed time t , we are integrating over all possible n -dimensional “polytopes” in the space of $t'_1, t'_2, \dots, t'_n$ defined by $t_0 < t'_1 < t'_2 < \dots < t'_n < t$.

Just as before, this does not capture the edges of the n -dimensional integration region where two or more of the time integration variables are equal ($t'_1 = t'_2$ or $t'_1 = t'_2 = \dots = t'_n$, etc), but again, we need not worry about these since these edge regions have measure zero in the n -dimensional integral. In fact, by inspection, one may guess that there should be a total of

$$\sum_{\ell=1}^{n-1} \binom{N}{\ell} \xrightarrow{N \rightarrow \infty} \frac{N^{(n-1)}}{(n-1)!} + \frac{N^{(n-2)}}{(n-2)!} + \dots + N \quad (35)$$

such “edge” contributions coming from picking $\mathcal{O}(\Delta t)^{n-\ell}$ terms from ℓ different $\hat{U}$ factors and $\mathcal{O}(\Delta t)^\ell$ terms from a single $\hat{U}$ factor. Clearly, these cannot compete with the leading $N^n/n!$ behavior, and hence vanish in the continuum limit.

Let us now collect all orders to find the *Dyson series*:

$$\boxed{\begin{aligned} \hat{U}(t, t_0) = & \hat{1} + \left(-\frac{i}{\hbar}\right) \int_{t_0}^t dt'_1 \hat{H}(t'_1) + \left(-\frac{i}{\hbar}\right)^2 \int_{t_0}^t dt'_2 \int_{t_0}^{t'_2} dt'_1 \hat{H}(t'_2) \hat{H}(t'_1) + \dots \\ & + \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^t dt'_n \int_{t_0}^{t'_n} dt'_{n-1} \dots \int_{t_0}^{t'_2} dt'_1 \hat{H}(t'_n) \hat{H}(t'_{n-1}) \dots \hat{H}(t'_1) + \dots \end{aligned}} \quad (36)$$

This looks quite daunting, so we introduce an important piece of notation: the *time-ordering operator* $\mathcal{T}$. The definition of time ordering is formally very straightforward:

$$\mathcal{T} \{ \hat{H}(t_1) \hat{H}(t_2) \} = \theta(t_1 - t_2) \hat{H}(t_1) \hat{H}(t_2) + \theta(t_2 - t_1) \hat{H}(t_2) \hat{H}(t_1) = \begin{cases} \hat{H}(t_1) \hat{H}(t_2), & \text{if } t_1 > t_2, \\ \hat{H}(t_2) \hat{H}(t_1), & \text{if } t_2 > t_1. \end{cases} \quad (37)$$

where $\theta(t)$ is the Heaviside step function defined as $\theta(t) = 1$ for $t > 0$ and $\theta(t) = 0$ for $t < 0$. The $t = 0$ case is again irrelevant, but may be set to $1/2$ to show how the time-ordering operator handles equal times symmetrically. A more precise and general definition is:

$$\mathcal{T} \{ \hat{H}(t_1) \dots \hat{H}(t_n) \} = \sum_{\pi \in S_n} \hat{H}(t_{\pi(1)}) \dots \hat{H}(t_{\pi(n)}) \theta(t_{\pi(1)} - t_{\pi(2)}) \dots \theta(t_{\pi(n-1)} - t_{\pi(n)}), \quad (38)$$

where the sum is over all permutations π in the symmetric group S_n on the set $\{1, 2, \dots, n\}$ (fancy way of saying: all possible arrangements of the numbers $1, 2, \dots, n$).

The ordering of the limits in the integral is very awkward, but by applying the time-ordering operator, one may symmetrize each nested integral and let $\mathcal{T}$ take care of the ordering for us:

$$\int_{t_0}^t dt'_n \int_{t_0}^{t'_n} dt'_{n-1} \dots \int_{t_0}^{t'_2} dt'_1 \hat{H}(t'_n) \dots \hat{H}(t'_1) = \frac{1}{n!} \int_{t_0}^t dt_1 \dots \int_{t_0}^t dt_n \mathcal{T} \{ \hat{H}(t_1) \dots \hat{H}(t_n) \}. \quad (39)$$

You can prove this equality by using the Heaviside. We will not do it in full generality here, but just illustrate the idea for $n = 2$.

In the case $n = 2$, the equality in Eq. (39) reads

$$\int_{t_0}^t dt_2 \int_{t_0}^{t_2} dt_1 \hat{H}(t_2) \hat{H}(t_1) = \frac{1}{2} \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \mathcal{T} \{ \hat{H}(t_1) \hat{H}(t_2) \},$$

where the limits of integration are decoupled and the time-ordering operator ensures the operators are ordered correctly. To prove this, we will start with the left-hand side and use the Heaviside step function to decouple the integration limits:

$$\int_{t_0}^t dt_2 \int_{t_0}^{t_2} dt_1 \hat{H}(t_2) \hat{H}(t_1) = \int_{t_0}^t dt_2 \int_{t_0}^t dt_1 \hat{H}(t_2) \hat{H}(t_1) \theta(t_2 - t_1) \quad (40)$$

Compare this with the right-hand side using the definition of the time-ordering operator:

$$\begin{aligned} \frac{1}{2} \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \mathcal{T} \{ \hat{H}(t_1) \hat{H}(t_2) \} \\ = \frac{1}{2} \int_{t_0}^t dt_1 \int_{t_0}^t dt_2 \left[\hat{H}(t_1) \hat{H}(t_2) \theta(t_1 - t_2) + \hat{H}(t_2) \hat{H}(t_1) \theta(t_2 - t_1) \right]. \end{aligned} \quad (41)$$

Splitting this into two integrals and renaming the dummy integration variables $t_1 \leftrightarrow t_2$ in the first term, we find that Eq. (40) and Eq. (41) are equal.

Substituting Eq. (39) into Eq. (36) gives the Dyson series in its more familiar form:

$$\hat{U}(t, t_0) = \sum_{n=0}^{\infty} \frac{1}{n!} \left(-\frac{i}{\hbar} \right)^n \int_{t_0}^t dt_1 \cdots \int_{t_0}^t dt_n \mathcal{T} \{ \hat{H}(t_1) \cdots \hat{H}(t_n) \} \quad (42)$$

$$= \left[\mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t dt' \hat{H}(t') \right] \right]. \quad (43)$$

Note how we pulled the time-ordering operator outside the sum, since it acts on each term in the same way. We then get the time-ordered Taylor expansion of the exponential function, a.k.a. the time-ordered exponential.

Examples to follow...